

# **MESS MANAGEMENT SYSTEM**

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for the partial fulfillment of the requirements to award the degree of

**Bachelor of Technology**

In

**Computer Science and Engineering**

**School of Engineering and Sciences**

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**[April, 2026]**

# Certificate

Date: 28 April, 2026

This is to certify that the project entitled **“MESS MANAGEMENT SYSTEM”** has been successfully carried out by **Aman Kumar Thakur (AP23110011682), Prithawi Kumar Jaiswal (AP23110011509), and Sameer Bachhar (AP23110011496)** under my supervision.

The work presented in this project is genuine, original, and has been completed as part of the academic requirements for the Full Stack Development course. To the best of my knowledge, this work has not been submitted elsewhere for the award of any degree or diploma.

This project is considered suitable for submission to **SRM University - AP** for the award of **Bachelor of Technology** in the **School of Engineering and Sciences**.

## Supervisor

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Mr. Ajay Yadav

Designation,

Affiliation.

# ACKNOWLEDGEMENT

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# 1. ABSTRACT

The **Mess Management System** is a web-based application developed to simplify and improve the daily operations of hostel mess facilities. In many institutions, mess management is still handled manually, which often leads to inefficiencies such as lack of menu transparency, unorganized feedback systems, and time-consuming attendance processes. This project aims to address these issues by providing a digital platform that connects students and mess administrators in a more efficient and structured manner.

The system allows students to view daily and weekly menus, mark their attendance digitally, provide ratings and feedback on meals, and raise complaints when necessary. On the other hand, administrators are provided with tools to manage menus, monitor student feedback, handle complaints, and analyze data for better decision-making. This creates a two-way communication system that improves overall management and user satisfaction.

The application is developed using a full stack approach, where the frontend is built using modern web technologies for a responsive and user-friendly interface, and the backend is designed to handle data processing, authentication, and API communication. The database is used to securely store user data, menu details, attendance records, and feedback information.

Security features such as authentication and protected routes ensure that only authorized users can access specific functionalities. The system also supports complete CRUD operations, allowing efficient data management across different modules.

Overall, the Mess Management System provides a practical solution to modernize traditional mess operations. It enhances transparency, reduces manual work, and improves communication between students and administrators. This project also demonstrates the application of full stack development concepts in solving real-world problems.

## 2. INTRODUCTION

The **Mess Management System** is a web-based application developed as part of our full stack development course with the aim of improving the efficiency of hostel mess operations. In many educational institutions, mess management is still handled through traditional methods, where information is either displayed manually or communicated verbally. This often leads to confusion, lack of proper coordination, and difficulty in maintaining records.

One of the common issues faced by students is the unavailability of timely information regarding the daily menu. Students usually visit the mess without knowing what food is being served, which can result in inconvenience and dissatisfaction. In addition, there is no proper system to collect and analyze feedback about food quality. Complaints are often informal and may not reach the concerned authorities effectively.

To address these challenges, the Mess Management System provides a centralized digital platform that connects students and administrators. The system allows students to access menu details, mark attendance, submit feedback, and raise complaints through an easy-to-use interface. At the same time, administrators can manage menus, monitor student activities, and respond to issues in a structured manner.

This project is developed using modern web technologies and follows a full stack approach. The frontend is responsible for user interaction, while the backend handles data processing, authentication, and communication with the database. The system ensures that all operations are performed efficiently and securely.

Through this project, we aim to demonstrate how technology can be used to solve real-world problems in a practical and effective way. It also provides valuable experience in designing, developing, and deploying a complete web application as part of our academic curriculum.

### 3. LITERATURE SURVEY

A literature survey involves studying existing systems and technologies used for managing library operations. Traditionally, libraries have relied on manual record-keeping systems where details of books, students, and transactions are maintained in registers. Although simple, these systems are inefficient, time-consuming, and prone to human errors such as incorrect entries, data loss, and difficulty in searching records.

With the advancement of technology, several computerized library systems have been developed. These systems store data digitally and provide faster access to information. However, many existing systems still lack advanced features such as proper user authentication, real-time tracking, and efficient database handling.

The proposed Online Library Management System improves upon existing systems by incorporating the following features:

- Automated book management system that allows easy addition, updating, and deletion of book records
- Secure login system for both admin and students to ensure controlled access
- Real-time tracking of issued and returned books to maintain accurate records
- Centralized database management using MySQL for efficient data storage and retrieval
- User-friendly interface developed using HTML, CSS, and Bootstrap for better user experience
- Reduction in manual workload and minimization of human errors
- Faster data processing and easy access to information

Compared to traditional and basic digital systems, the proposed system provides better performance, improved accuracy, enhanced security, and scalability. It ensures efficient handling of library operations and offers a reliable solution suitable for modern educational institutions.

## 4. OBJECTIVES

The main objective of developing the **Mess Management System** is to provide an efficient and reliable digital solution for managing hostel mess operations. The system is designed to reduce manual work, improve communication, and ensure better coordination between students and mess administrators.

**The specific objectives of the project are as follows:**

- To develop a centralized platform where students can easily access daily and weekly mess menus without any confusion or delay.
- To replace the traditional manual attendance system with a digital method that allows students to mark their attendance quickly and accurately.
- To create a structured feedback system where students can rate meals and provide comments, helping administrators understand food quality and preferences.
- To provide a proper complaint management system where students can raise issues and track their status, ensuring transparency and accountability.
- To enable administrators to manage menus, monitor student activities, and handle complaints through a dedicated dashboard.
- To implement secure user authentication so that only authorized users can access the system and perform specific operations.
- To support full CRUD (Create, Read, Update, Delete) operations for efficient data management across different modules.
- To design a user-friendly and responsive interface that can be accessed easily on different devices.
- To demonstrate the practical implementation of full stack development concepts as part of academic learning.

Overall, the objective of this project is to improve the existing mess management process by introducing a modern, digital, and efficient system.



## 5. PROBLEM STATEMENT

In many hostels and educational institutions, the management of mess activities is still carried out using traditional and manual methods. This leads to several challenges that affect both students and mess administrators.

One of the major issues is the lack of proper communication regarding the daily menu. Students often visit the mess without knowing what food is being served, which can result in inconvenience and dissatisfaction. There is no centralized system where students can easily check menu details in advance.

Another significant problem is the absence of a structured feedback mechanism. Students may have opinions about food quality, quantity, or hygiene, but there is no proper platform to record and analyze this feedback. As a result, administrators are unable to make informed decisions to improve services.

Attendance management is also handled manually in many cases, which is time-consuming and prone to errors. Maintaining records on paper or through informal methods makes it difficult to track and manage attendance efficiently.

In addition, there is no organized system for handling complaints. Students may raise issues verbally or through informal channels, but these complaints are often not properly recorded or resolved in a timely manner.

Due to the lack of digital systems, data is not stored in a structured format, making it difficult to analyze trends, monitor performance, or improve overall mess operations.

Therefore, there is a need for a reliable and efficient system that can address these problems by providing better communication, proper record management, and improved interaction between students and mess administration.

## 6. TECHNOLOGY STACK

The **Mess Management System** is developed using a full stack approach, combining modern technologies for both frontend and backend development. Each technology is selected to ensure performance, scalability, and ease of use.

### Frontend Technologies

The frontend of the application is responsible for user interaction and interface design.

- **React.js**  
React is used to build a dynamic and responsive user interface. It allows the application to be divided into reusable components, making development more efficient and organized.
- **Vite**  
Vite is used as a build tool and development server. It provides faster loading and improves development speed compared to traditional tools.
- **React Router**  
React Router is used for navigation between different pages such as login, dashboard, and menu pages without reloading the application.
- **Axios**  
Axios is used to communicate with the backend APIs. It helps in sending and receiving data efficiently between frontend and backend.
- **CSS/ Tailwind CSS**  
Styling is handled using CSS and utility-based frameworks to create a clean, responsive, and user-friendly interface.

### Backend Technologies

The backend handles business logic, data processing, and communication with the database.

- **Node.js**  
Node.js is used as the runtime environment for executing server-side JavaScript code.
- **Express.js**  
Express is used to build RESTful APIs. It simplifies server creation and helps in handling requests and responses efficiently.

- **MongoDB**

MongoDB is a NoSQL database used to store application data in a flexible and structured format.

- **Mongoose**

Mongoose is an Object Data Modeling (ODM) library used to define schemas and interact with MongoDB in an organized way.

- **JSON Web Token (JWT)**

JWT is used for authentication and authorization. It ensures that only valid users can access protected routes.

- **bcrypt**

bcrypt is used to securely hash user passwords before storing them in the database.

## **Tools and Platforms**

- **Postman**

Used for testing APIs and verifying backend functionality.

- **Git and GitHub**

Used for version control and project management.

- **Vercel**

Used for deploying the frontend application.

- **Render**

Used for deploying the backend server.

- **MongoDB Atlas**

Used as a cloud-based database service to store and manage application data.

## **Summary**

The combination of these technologies ensures that the system is efficient, scalable, and secure. The use of the MERN stack allows seamless integration between frontend and backend, making the application suitable for real-world use.

## 7. SYSTEM ARCHITECTURE

The **Mess Management System** follows a three-tier architecture that separates the application into different layers. This structure helps in maintaining the system efficiently and allows smooth communication between different components.

The architecture consists of the following layers:

### 1. Presentation Layer (Frontend)

The presentation layer is the user interface of the application, developed using React.js. It is responsible for interacting with users and displaying information in a clear and user-friendly manner.

Students and administrators access the system through this layer. It includes pages such as login, dashboard, menu display, attendance, and complaint sections. The frontend sends requests to the backend and displays the responses received.

### 2. Application Layer (Backend)

The application layer is developed using Node.js and Express.js. This layer acts as the core of the system and handles all business logic.

It is responsible for:

- Processing user requests
- Validating data
- Handling authentication using JWT
- Managing API routes
- Communicating with the database

When a user performs an action (such as login or viewing the menu), the request is sent to the backend, where it is processed and the appropriate response is generated.

### 3. Data Layer (Database)

The data layer uses MongoDB as the database to store all application data. Mongoose is used to define schemas and interact with the database.

This layer stores:

- User information
- Menu details
- Attendance records
- Ratings and feedback
- Complaints

The backend communicates with the database to store, retrieve, update, and delete data as required.

### **Working of the System**

1. The user interacts with the frontend (React application).
2. The frontend sends HTTP requests to the backend APIs.
3. The backend processes the request and interacts with the database if needed.
4. The database returns the required data to the backend.
5. The backend sends a response back to the frontend.
6. The frontend displays the result to the user.

### **Advantages of This Architecture**

- Clear separation of frontend, backend, and database
- Easy to maintain and update
- Scalable for future improvements
- Secure handling of user data and authentication

Overall, the system architecture ensures smooth communication between all components and supports efficient performance of the application.

## 8. MODULES DESCRIPTION

The **Mess Management System** is divided into different modules to organize the functionality of the application in a structured manner. Each module is designed to handle specific tasks, making the system efficient and easy to manage.

### 1. User Authentication Module

This module is responsible for managing user access to the system. It includes features such as registration and login for both students and administrators.

- New users can register by providing their details.
- Existing users can log in using their credentials.
- Authentication is handled securely using JWT.
- Only authorized users can access protected sections of the system.

### 2. Student Module

The student module provides all the functionalities required by students.

- View daily and weekly menus
- Mark attendance digitally
- Rate meals and provide feedback
- Submit complaints
- View personal activity history

This module ensures that students can interact with the mess system easily and efficiently.

### 3. Admin Module

The admin module is designed for mess administrators to manage the system.

- Admin login
- Manage student accounts (approve or monitor users)
- Create and update menus
- View and analyze ratings
- Handle complaints and update their status

This module helps administrators maintain control over the system and improve service quality.

### 4. Menu Management Module

This module allows administrators to manage mess menus.

- Add new menu for specific dates
- Update existing menu items
- Display menu to students

It ensures that menu information is always updated and accessible.

## **5. Attendance Module**

The attendance module manages the attendance of students for different meals.

- Students can mark attendance for breakfast, lunch, snacks, and dinner
- Records are stored digitally
- Admin can view attendance reports

This replaces manual attendance methods and reduces errors.

## **6. Rating and Feedback Module**

This module allows students to provide feedback on meals.

- Students can rate meals (e.g., 1–5 stars)
- Comments can be added for detailed feedback
- Admin can view and analyze ratings

This helps improve food quality based on user feedback.

## **7. Complaint Management Module**

This module handles complaints raised by students.

- Students can submit complaints
- Admin can view, respond, and update complaint status
- Proper tracking of issues is maintained

This ensures transparency and better communication.

## **8. API Module**

This module manages communication between frontend and backend.

- Handles all API requests (GET, POST, PUT, DELETE)
- Ensures smooth data transfer
- Supports integration between different modules

## 9. DATABASE DESIGN

The **Mess Management System** uses MongoDB as the primary database for storing and managing application data. MongoDB is a NoSQL database that stores data in a flexible, JSON-like format, making it suitable for modern web applications.

To interact with the database in an organized manner, Mongoose is used as an Object Data Modeling (ODM) library. It allows defining schemas and ensures structured data handling within the application.

### **Collections Used in the System**

The database is divided into multiple collections, each responsible for storing specific types of data.

#### **1. Users Collection**

This collection stores information about both students and administrators.

Fields include:

- Name
- Email
- Password (hashed)
- Roll Number
- Hostel Name
- Room Number
- Phone Number
- Role (Student/Admin)
- Verification Status

This collection is used for authentication and user management.

#### **2. Menus Collection**

This collection stores daily and weekly menu details.

Fields include:

- Date
- Day
- Breakfast Items
- Lunch Items
- Snacks Items



- Dinner Items

This allows students to view meal information in advance.

### **3. Ratings Collection**

This collection stores feedback provided by students for different meals.

Fields include:

- User ID
- Menu ID
- Meal Type
- Rating (1-5)
- Comment

This helps in analyzing food quality and student satisfaction.

### **4. Attendance Collection**

This collection stores attendance records of students.

Fields include:

- User ID
- Date
- Breakfast Attendance
- Lunch Attendance
- Snacks Attendance
- Dinner Attendance

This replaces manual attendance tracking and improves accuracy.

### **5. Complaints Collection**

This collection stores complaints raised by students.

Fields include:

- User ID
- Title
- Description
- Status (Pending/Resolved)
- Admin Response

This helps in tracking and resolving issues efficiently.

### **Database Relationships**

Although MongoDB is a NoSQL database, relationships are maintained using references such as User ID and Menu ID. These references help in linking data across different collections.

### **Advantages of Using MongoDB**

- Flexible schema design
- Easy to store complex data
- Scalable for large applications
- Faster data retrieval
- Suitable for real-time applications

## 10. AUTHENTICATION AND SECURITY

Security is an essential aspect of the **Mess Management System**, as it involves handling sensitive user data such as login credentials, personal information, and system operations. Proper authentication and security measures are implemented to ensure that the system is safe and accessible only to authorized users.

### User Authentication

The system uses **JSON Web Token (JWT)** for authentication. When a user logs in with valid credentials, a token is generated by the backend and sent to the frontend. This token is then used to verify the identity of the user for future requests.

- Users must log in to access the system
- After successful login, a JWT token is generated
- The token is stored securely on the client side
- Each request to protected routes includes the token

This approach ensures that only authenticated users can access specific features of the application.

### Authorization

Authorization is implemented to control access to different parts of the system.

- Students can access only student-related features
- Administrators have access to management features
- Protected routes are implemented using middleware

This ensures that users can only perform actions that are permitted based on their role.

### Password Security

To protect user passwords, the system uses **bcrypt** for hashing.

- Passwords are never stored in plain text
- Before saving to the database, passwords are encrypted
- During login, entered passwords are compared with hashed values

This ensures that even if data is exposed, passwords remain secure.

### Protected Routes

Certain routes in the backend are protected using authentication middleware.

- Only logged-in users can access private APIs
- Token verification is performed before processing requests

- Unauthorized access is restricted

This prevents misuse of system functionalities.

### **Environment Variables**

Sensitive information such as database connection strings and secret keys are stored using environment variables.

- JWT secret keys are stored securely
- Database credentials are not exposed in code
- Improves overall security of the application

### **Data Security**

The system ensures secure handling of data through:

- Proper validation of user inputs
- Error handling to avoid system crashes
- Controlled access to database operations

## 11. CRUD OPERATIONS

The **Mess Management System** supports full CRUD operations, which are essential for managing data efficiently within the application. CRUD stands for Create, Read, Update, and Delete, and these operations are implemented across different modules of the system.

### 1. Create Operation

The Create operation is used to add new data into the system.

Examples:

- Registering a new student
- Adding a new menu by the admin
- Submitting a complaint
- Providing ratings and feedback

This operation allows users and administrators to input new information into the database.

### 2. Read Operation

The Read operation is used to retrieve and display data from the system.

Examples:

- Viewing daily and weekly menus
- Checking attendance records
- Viewing submitted complaints
- Displaying ratings and feedback

This operation ensures that users can access the required information easily.

### 3. Update Operation

The Update operation is used to modify existing data.

Examples:

- Updating menu details by the admin
- Changing complaint status (Pending to Resolved)
- Editing user information
- Updating ratings or feedback

This helps keep the data accurate and up to date.

#### **4. Delete Operation**

The Delete operation is used to remove data from the system.

Examples:

- Deleting outdated menu entries
- Removing incorrect records
- Deleting user data if required
- Removing unnecessary complaints or feedback

This operation helps maintain data cleanliness and relevance.

#### **Implementation in the System**

CRUD operations are implemented using RESTful APIs in the backend. Different HTTP methods are used:

- POST → Create
- GET → Read
- PUT → Update
- DELETE → Delete

These operations are connected to the database through Mongoose, ensuring smooth data handling.

## 12. API IMPLEMENTATION

The **Mess Management System** uses RESTful APIs to enable communication between the frontend and backend. These APIs are developed using Node.js and Express.js and are responsible for handling user requests, processing data, and interacting with the database.

Each API endpoint is designed to perform specific operations such as authentication, data retrieval, and data management. The APIs follow standard HTTP methods to ensure a structured and organized approach.

### Types of API Requests

The system uses the following HTTP methods:

- **POST** → Used to create new data (e.g., registration, login, adding menu)
- **GET** → Used to fetch data (e.g., menu, ratings)
- **PUT** → Used to update existing data
- **DELETE** → Used to remove data

### Authentication APIs

These APIs are used for user login and registration.

- **POST /api/auth/register**  
Used to register a new student by providing required details.
- **POST /api/auth/login**  
Used for student login and authentication.
- **POST /api/auth/admin/login**  
Used for administrator login.

These APIs generate JWT tokens after successful authentication.

### Menu APIs

These APIs manage menu-related operations.

- **GET /api/menu/today**  
Fetches the menu for the current day.
- **GET /api/menu**  
Retrieves all available menu records.
- **POST /api/admin/menu**  
Allows the admin to create or update menu details.

## Rating APIs

These APIs handle meal ratings and feedback.

- **POST /api/ratings**  
Used by students to submit ratings and comments.
- **GET /api/ratings/all**  
Used by admin to view all ratings.
- **GET /api/ratings/meal/:menuId:mealType**  
Fetches ratings for a specific meal.

## Attendance APIs

These APIs manage attendance records.

- **POST /api/attendance**  
Used to mark attendance.
- **GET /api/attendance/today**  
Retrieves attendance for the current day.

## Complaint APIs

These APIs manage complaints submitted by students.

- **POST /api/complaints**  
Used to submit a complaint.
- **GET /api/complaints**  
Retrieves complaints for the user.
- **PUT /api/admin/complaints/:id**  
Used by admin to update complaint status.

## API Workflow

1. The frontend sends a request to a specific API endpoint.
2. The backend processes the request using controllers.
3. The backend interacts with the database using Mongoose.
4. The response is sent back to the frontend in JSON format.
5. The frontend displays the result to the user.

## Security in APIs

- JWT tokens are used for protected routes



- Authorization headers are required for sensitive operations
- Input validation is applied to prevent errors

### **Summary**

The API implementation ensures smooth communication between different parts of the system. It allows efficient data handling, secure operations, and proper integration between frontend and backend components.

## 13. API TESTING USING POSTMAN

To ensure that all backend functionalities of the **Mess Management System** work correctly, API testing was performed using Postman. Postman is a widely used tool that allows developers to send HTTP requests and analyze responses from the server. A structured Postman collection was created to organize and test all the API endpoints of the system. This helped in verifying that each API performs the expected operation and returns correct responses.

### Purpose of API Testing

The main purpose of using Postman for testing was:

- To verify that all API endpoints are working correctly
- To check request and response data
- To ensure proper integration between frontend and backend
- To identify and fix errors during development

### APIs Tested

The following APIs were tested using Postman:

#### Authentication APIs

- **POST /api/auth/register** → Used for student registration
- **POST /api/auth/login** → Used for student login
- **POST /api/auth/admin/login** → Used for admin login

These APIs were tested by sending user data in JSON format and verifying successful login responses and token generation.

#### Menu APIs

- **GET /api/menu/today** → Fetch today's menu
- **GET /api/menu** → Fetch all menus

These APIs were tested to ensure correct retrieval of menu data from the database.

#### Rating APIs

- **POST /api/ratings** → Submit meal rating
- **GET /api/ratings/all** → Retrieve all ratings

These APIs were tested to confirm that feedback is properly stored and fetched.

#### Admin APIs

- **POST /api/admin/menu** → Create or update menu

This API was tested to verify that administrators can manage menu data successfully.

### **Authentication Testing**

For protected routes, JWT tokens were used during testing. After login, the token received was added to the request header.

Example:

Authorization: Bearer

This ensured that only authorized users could access secure endpoints.

### **Testing Process**

1. API endpoint was selected in Postman
2. Request method (GET, POST, etc.) was chosen
3. Required data was entered in the request body
4. Request was sent to the server
5. Response was analyzed for correctness

### **Benefits of Using Postman**

- Easy and efficient API testing
- Helps in debugging errors
- Ensures proper backend functionality
- Validates data flow between frontend and backend

### **Summary**

All APIs of the system were successfully tested using Postman. This ensured that the backend services are reliable, secure, and functioning as expected, which contributes to the overall performance of the application.

## 14. USER INTERFACE DESIGN

The **User Interface (UI)** of the Mess Management System is designed to be simple, clean, and user-friendly so that users can easily interact with the system without any difficulty. The focus of the design is to provide a smooth experience for both students and administrators.

The frontend of the application is developed using React.js, which allows the creation of reusable components and ensures a responsive design across different devices such as desktops, tablets, and mobile phones.

### Design Principles

The following principles were considered while designing the user interface:

- **Simplicity:** The interface is kept simple to ensure ease of use for all users.
- **Consistency:** Similar design patterns are used across all pages for better usability.
- **Responsiveness:** The application adapts to different screen sizes.
- **Accessibility:** Important features are easily accessible without complex navigation.

### Main Pages of the System

The system consists of multiple pages designed for different functionalities:

#### 1. Login and Registration Page

- Allows users to create an account and log in
- Simple form layout with input validation
- Clear instructions for users

#### 2. Dashboard

- Displays key information such as menu, attendance, and quick actions
- Provides easy navigation to different sections
- Acts as the main interface after login

#### 3. Menu Page

- Displays daily and weekly menus
- Organized layout for different meals (breakfast, lunch, snacks, dinner)
- Easy to read and understand

#### 4. Attendance Page

- Allows students to mark attendance with a single click
- Shows attendance status clearly

### **5. Ratings and Feedback Page**

- Allows users to rate meals and provide comments
- Displays previous feedback

### **6. Complaint Page**

- Enables users to submit complaints
- Shows status of submitted complaints

### **7. Admin Panel**

- Provides administrative controls
- Allows menu management, complaint handling, and data monitoring
- Organized layout for efficient management

### **UI Components**

The interface uses reusable components such as:

- Buttons
- Forms
- Cards
- Navigation bars
- Modals

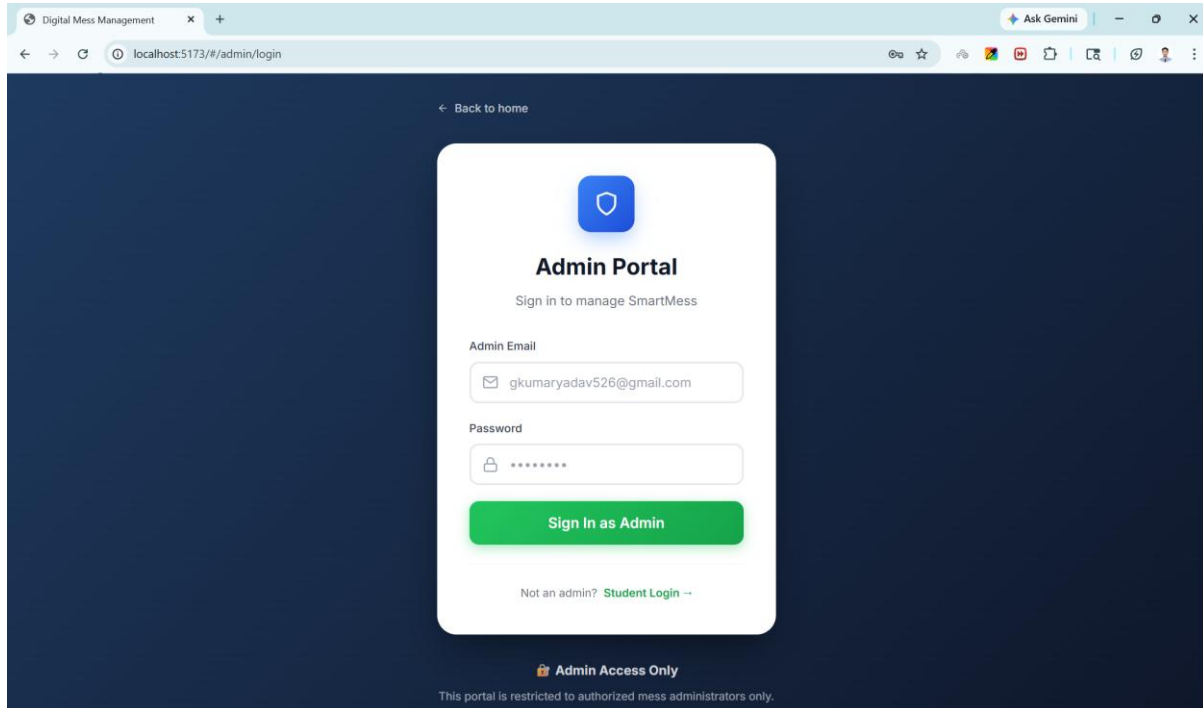
These components help maintain consistency and improve development efficiency.

### **Summary**

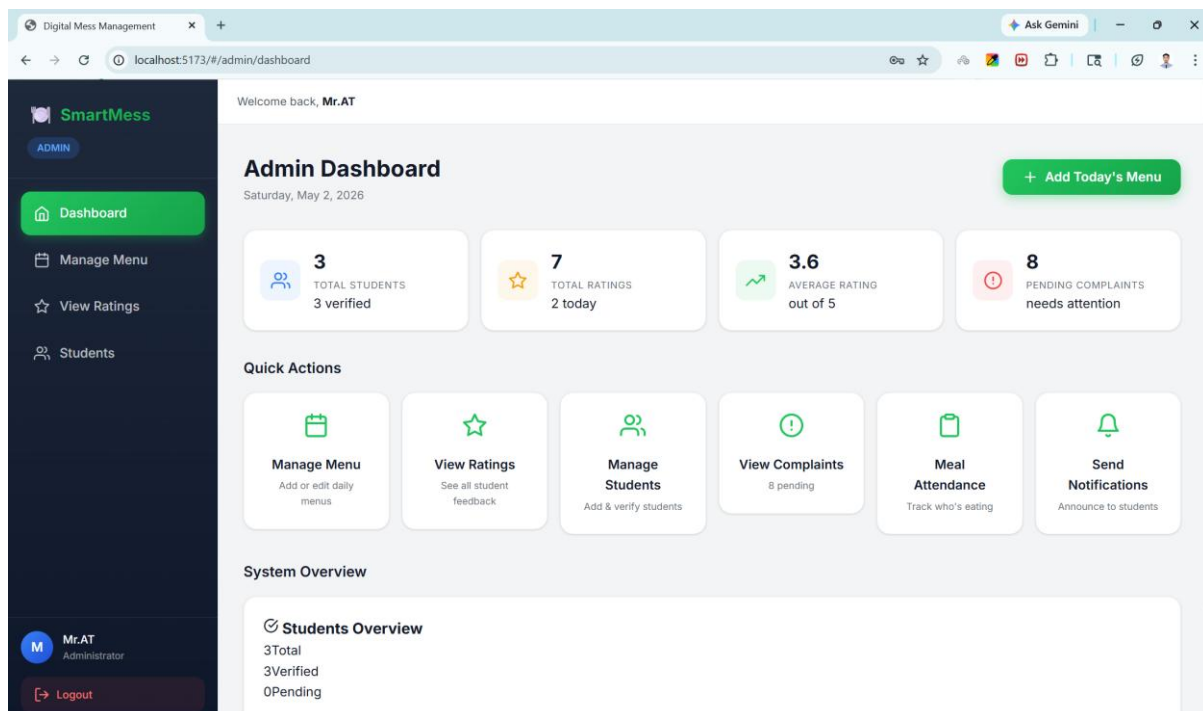
The user interface is designed to provide a smooth and efficient user experience. It ensures easy navigation, quick access to features, and clear presentation of information, making the system practical and user-friendly.

# 15. OUTPUT (SCREENSHOTS)

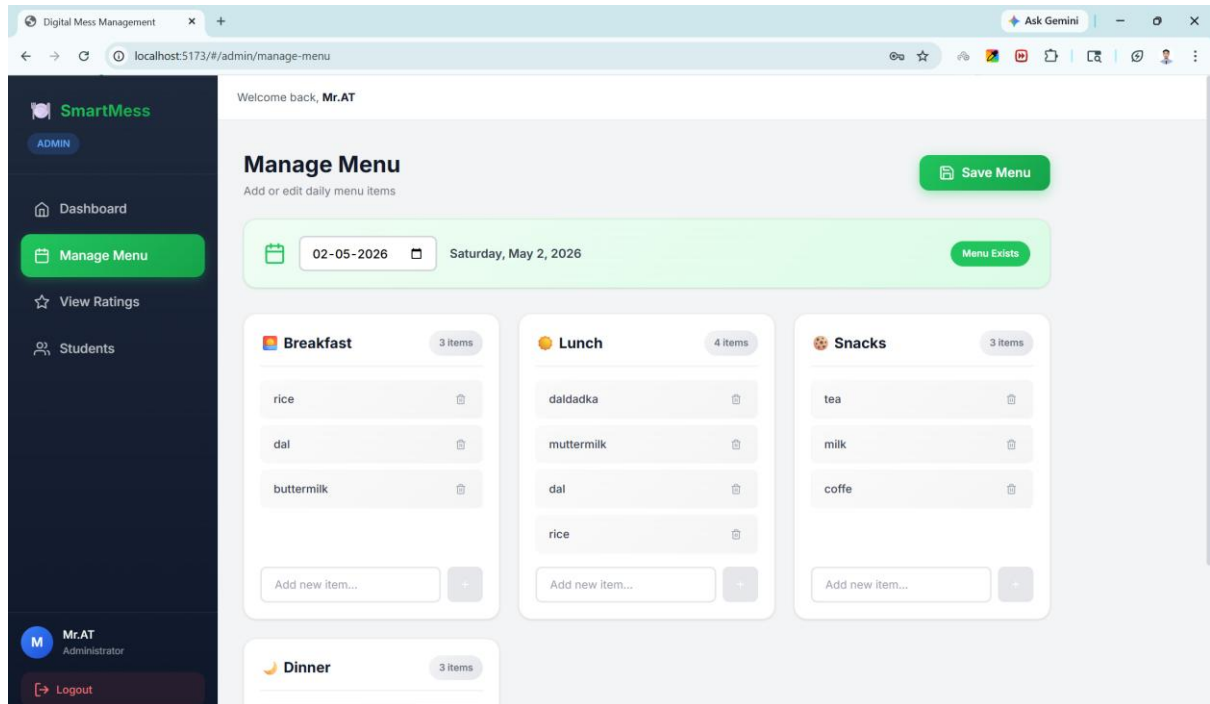
## 1. Login Page Admin



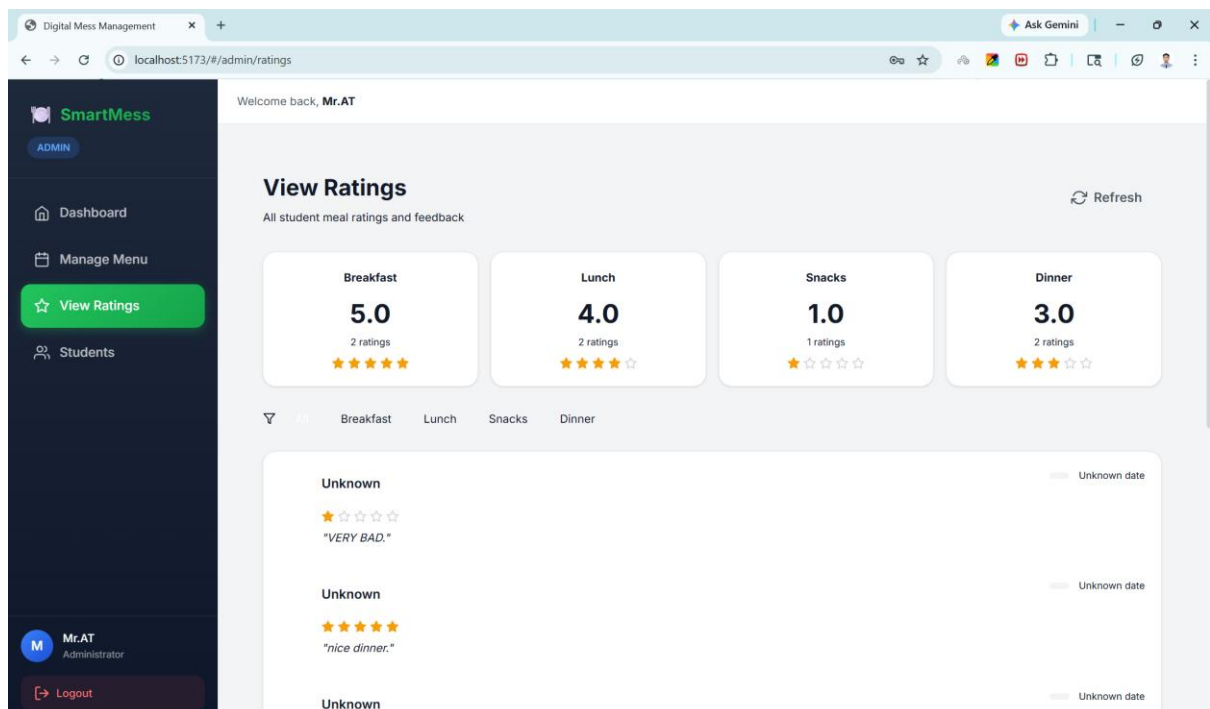
## 2. Admin Dashboard



### 3. Manage Menu



### 4. View Ratings



## 5. Manage Students

Welcome back, **Mr.AT**

### Manage Students

Add, verify, and manage student accounts

Refresh Add Student

Search by name, roll numbe

All (3) Pending Approval (3) Approved

| STUDENT                                                                                                                           | ROLL NUMBER     | HOSTEL         | STATUS             | RATINGS   | ACTIONS                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>raja ram</b><br>er.thakuramankumar@gmail.com | AP2311001133333 | GANGA - 102    | Approved<br>Active | 1 ratings |   |
|  <b>Aman Kumar Thakur</b><br>amankumar@gmail.com | AK11            | AR12 - V101    | Approved<br>Active | 0 ratings |   |
|  <b>Aman Kumar</b><br>thakuramankumar@gmail.com  | AP23110011682   | Vedavati - 101 | Approved<br>Active | 4 ratings |   |

**Mr.AT**  
Administrator  
Logout

## 6. Add New Student

### Add New Student

Full Name

Email

Roll Number

Phone Number

Hostel Name

Room Number

Password

Cancel Add Student



## 7. View Complaints

Digital Mess Management

Welcome back, Mr.AT

### View Complaints

Manage and respond to student complaints

Refresh

8 Total

8 Pending

0 In Progress

0 Resolved

Unknown

Service

THE SERVICE OF STAFF IS VERY BAD IN THE MESS.

May 2, 2026 at 8:02 PM

Pending

Response

Unknown

Food\_quality

the food quality is not good it is too bad and not in a well proper mentailed.

May 2, 2026 at 7:57 PM

Pending

Response

## 8. Meal Attendance

Digital Mess Management

Welcome back, Mr.AT

### Meal Attendance

Track how many students will attend each meal

Refresh

Breakfast

0 Present, 0 Absent, 3 Not Marked

Lunch

0 Present, 0 Absent, 3 Not Marked

Snacks

2 Present, 0 Absent, 1 Not Marked

66.7% Attendance

Dinner

0 Present, 0 Absent, 3 Not Marked

Saturday, May 2, 2026

Today

All Meals

Breakfast

Lunch

Snacks

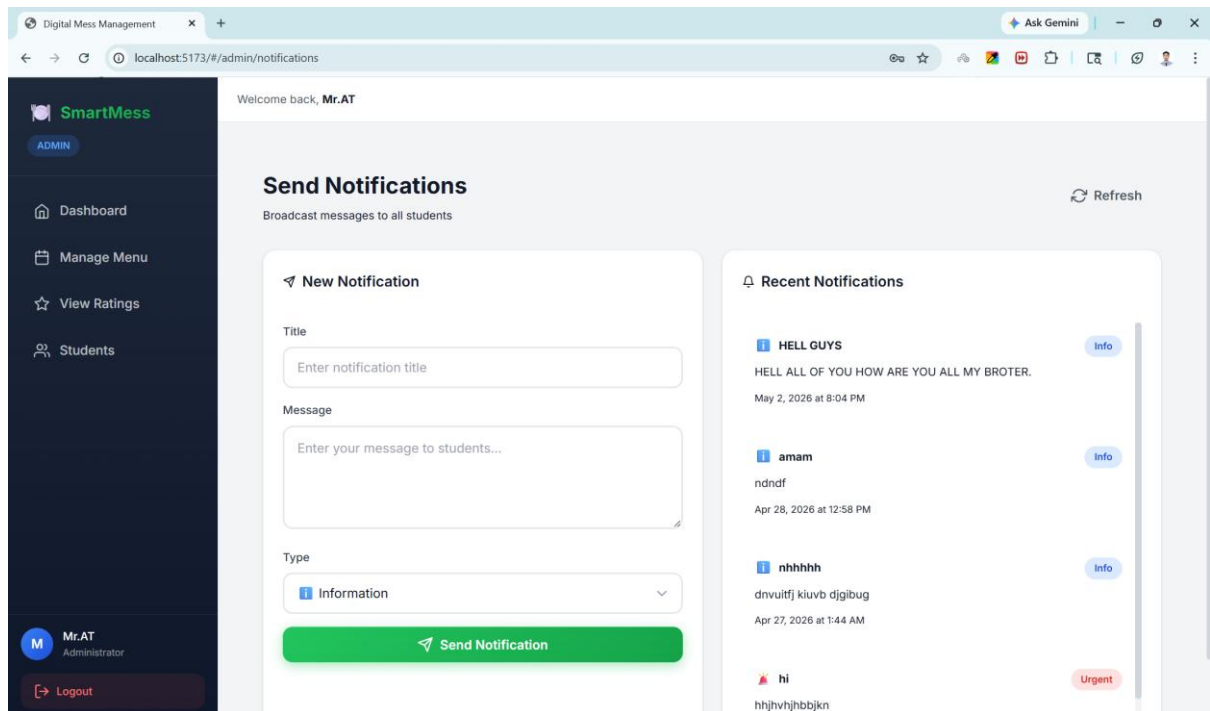
Dinner

#### Student Attendance

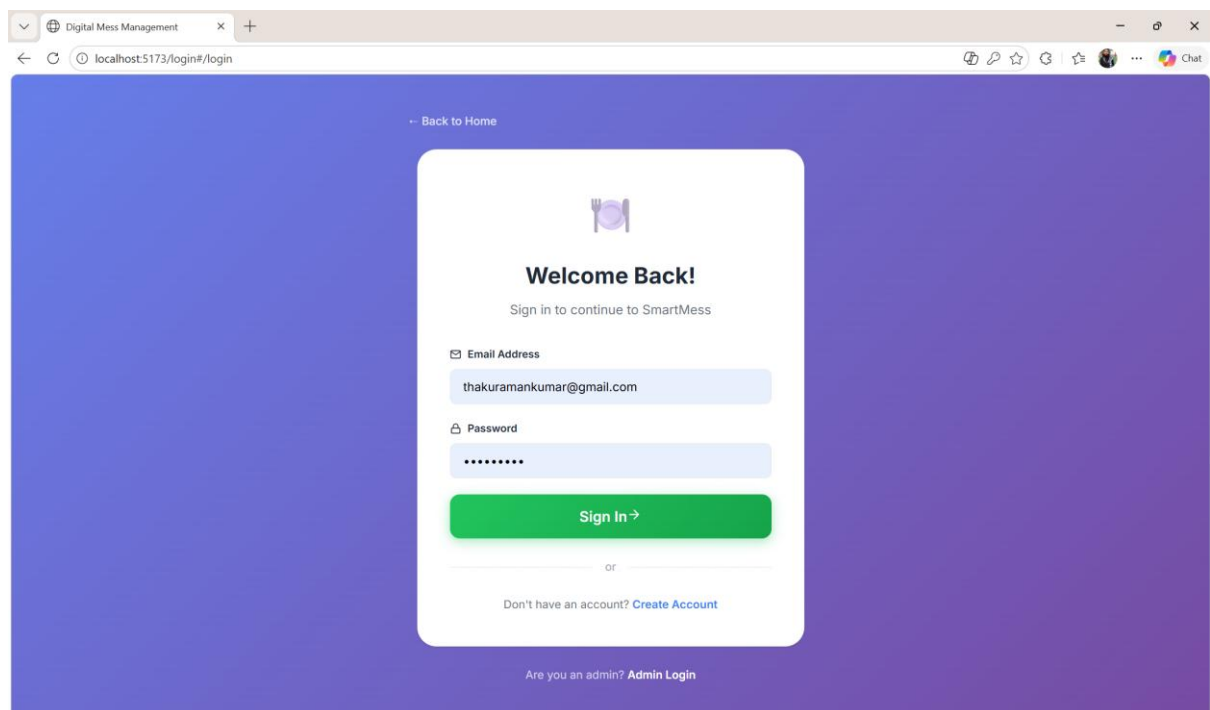
3 students

| STUDENT           | ROLL NUMBER | HOSTEL | SNACKS       |
|-------------------|-------------|--------|--------------|
| Aman Kumar        |             | -      | ✓ Present    |
| Aman Kumar Thakur |             | -      | - Not Marked |

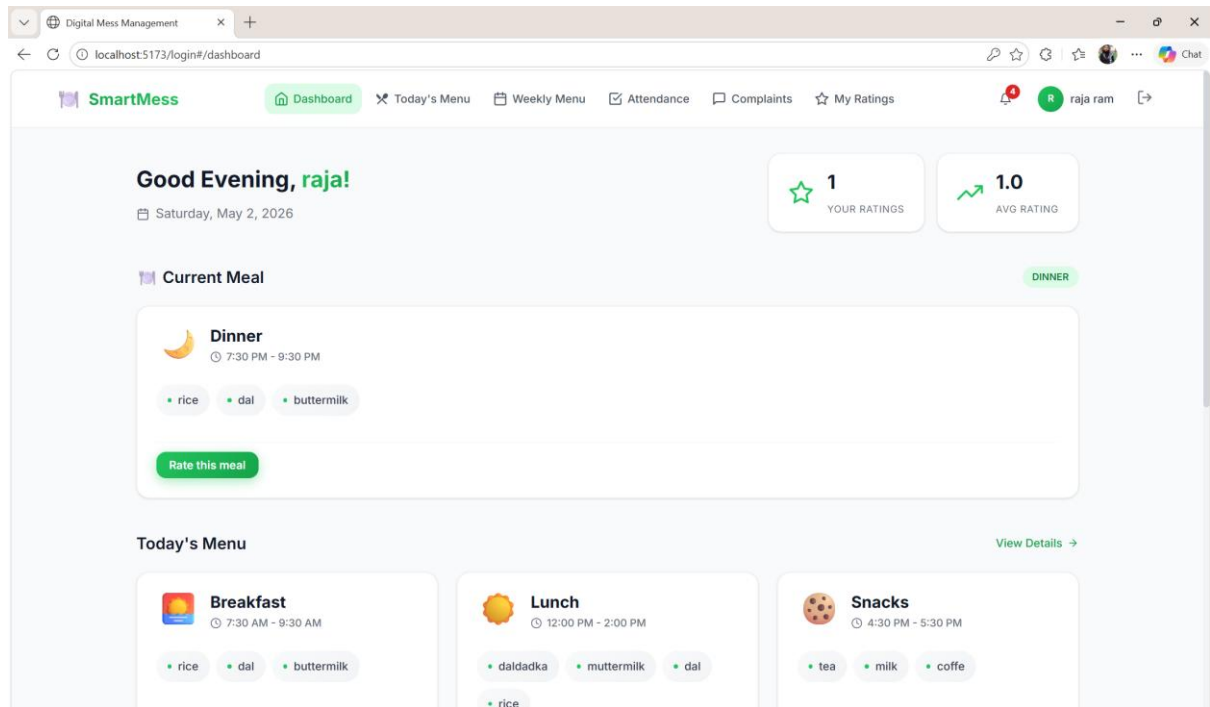
## 9. Send Notifications



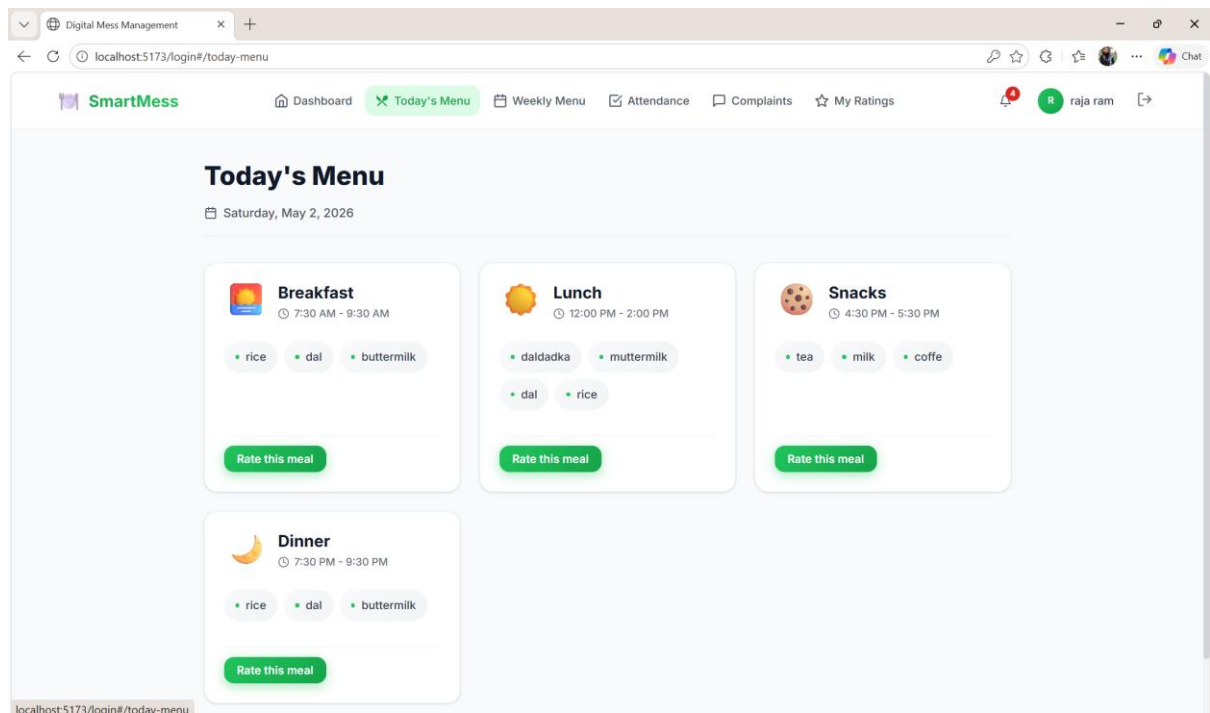
## 10. Login Page User



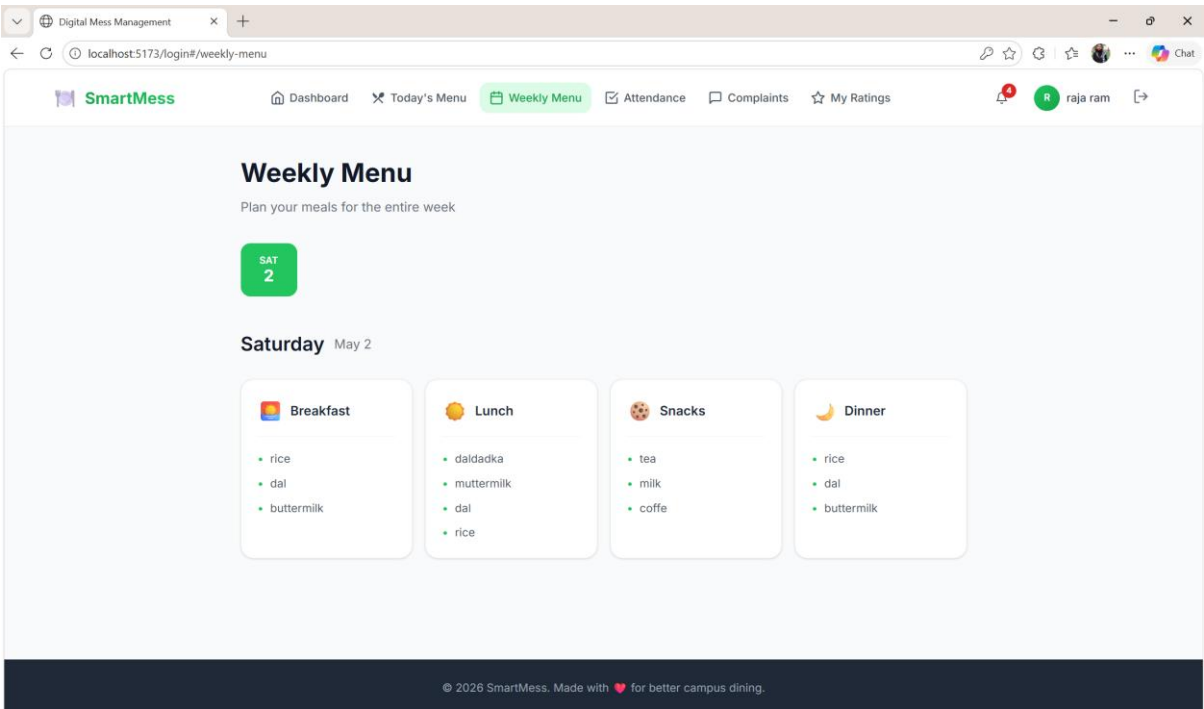
## 11. User Dashboard



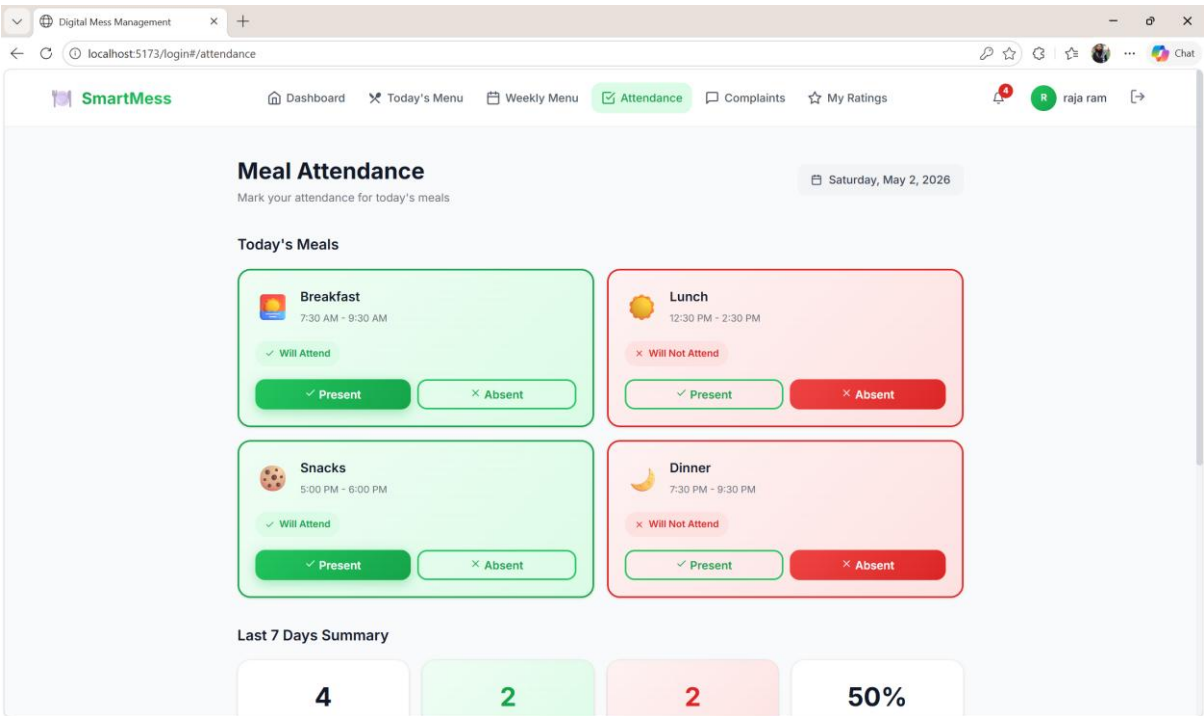
## 12. Today's Menu



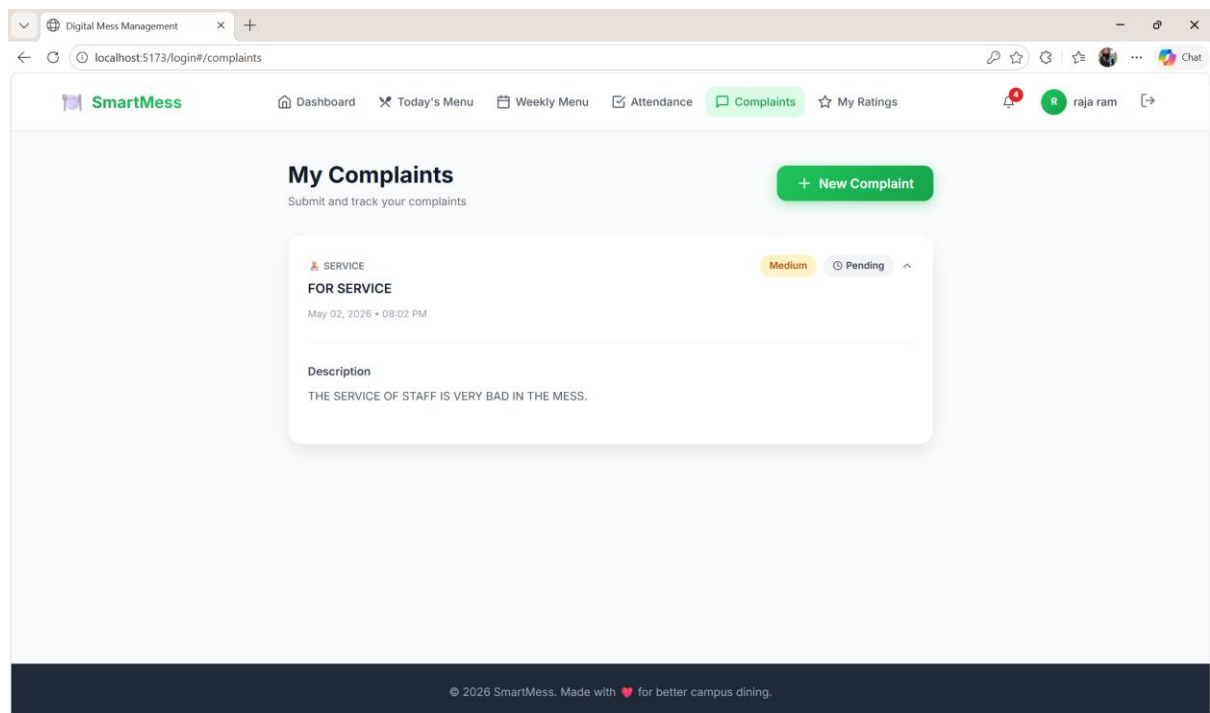
## 13. Weekly Menu



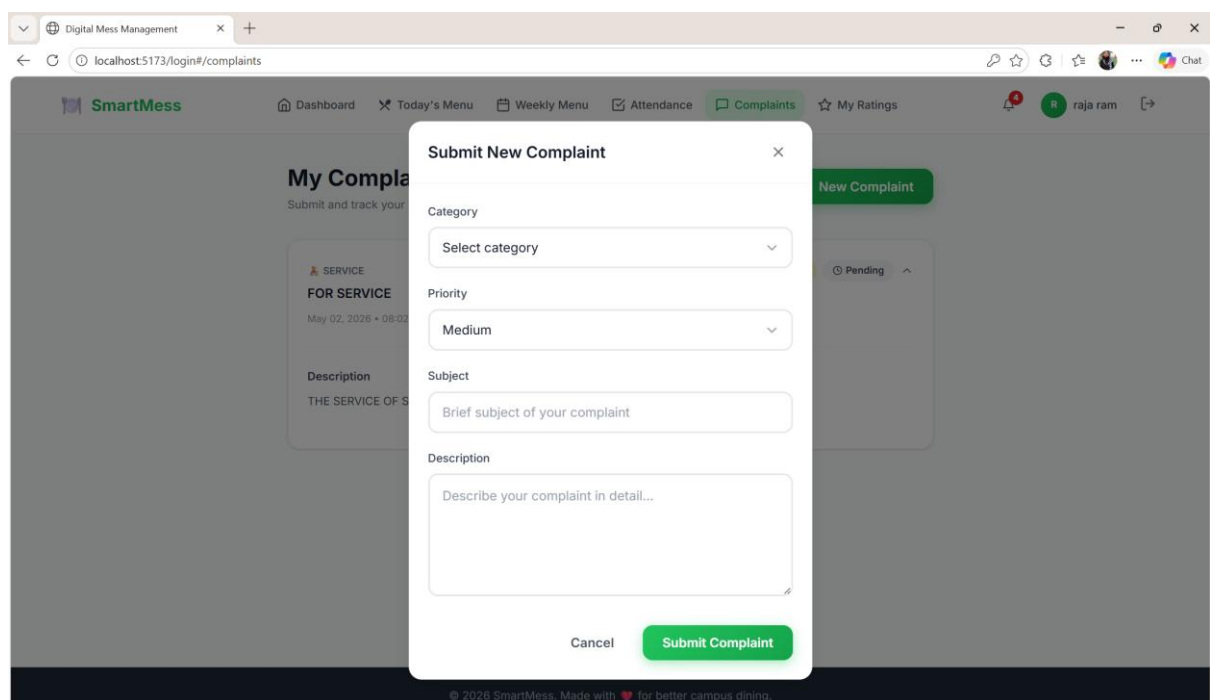
## 14. Meal Attendance



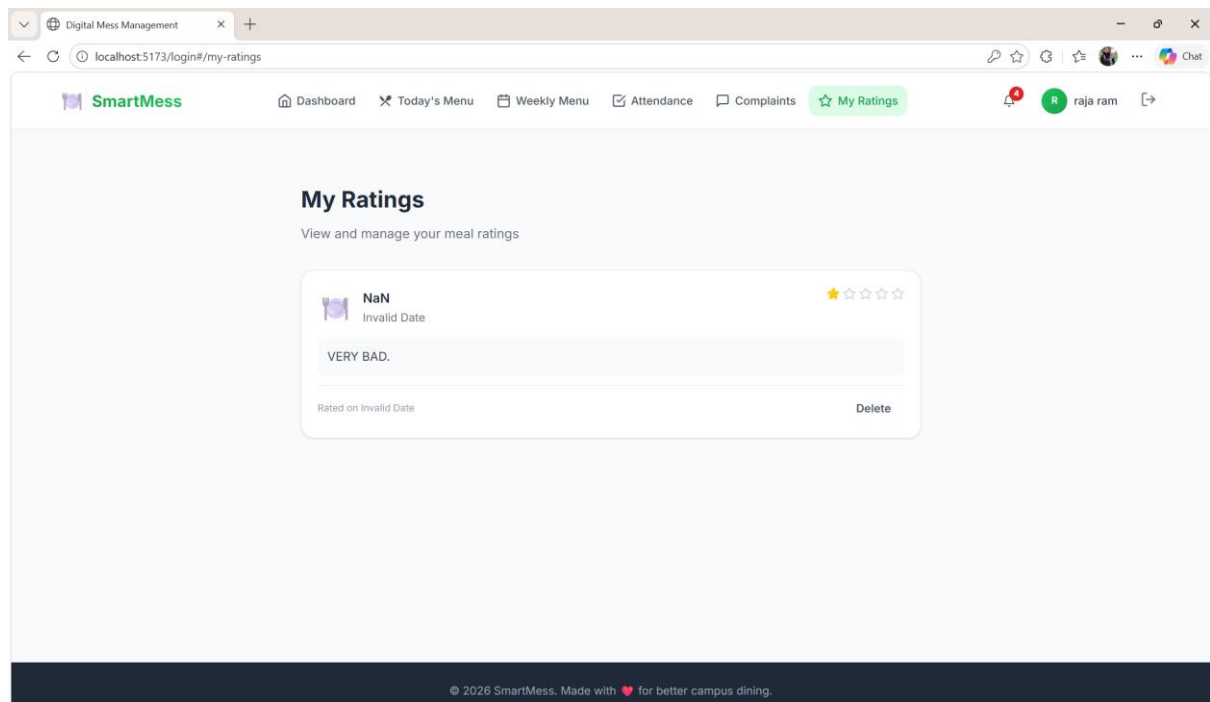
## 15. My Complaints



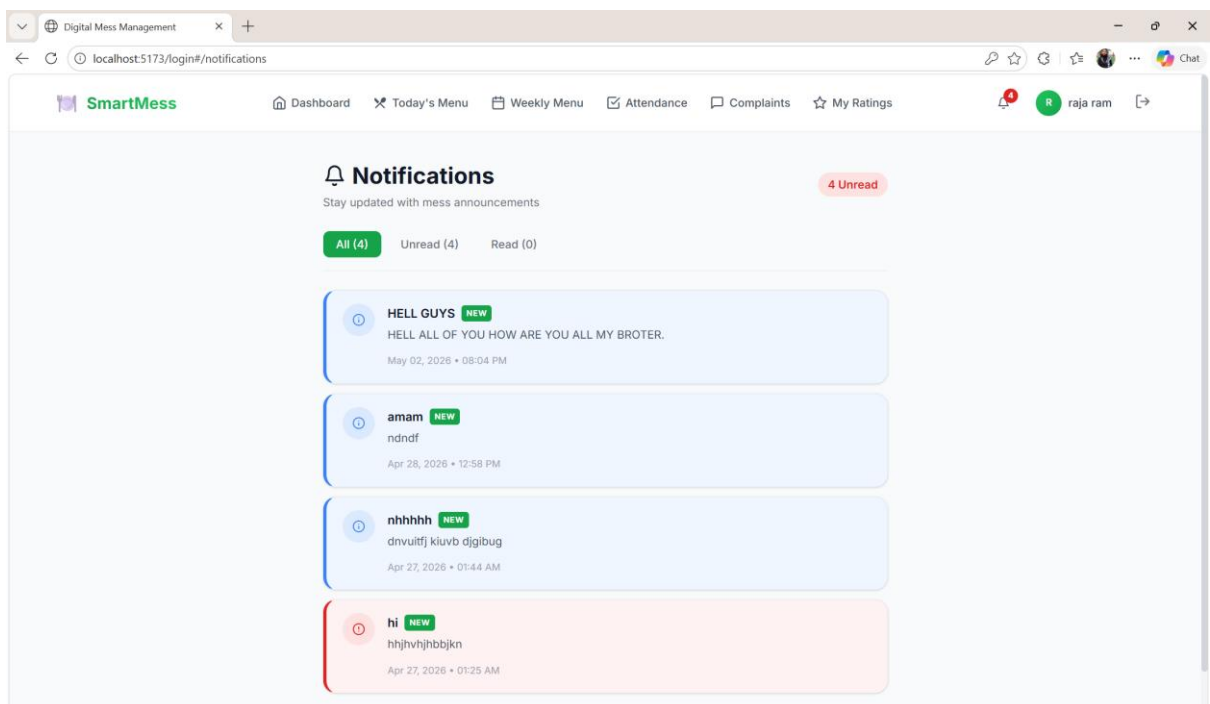
## 16. Submit New Complaint



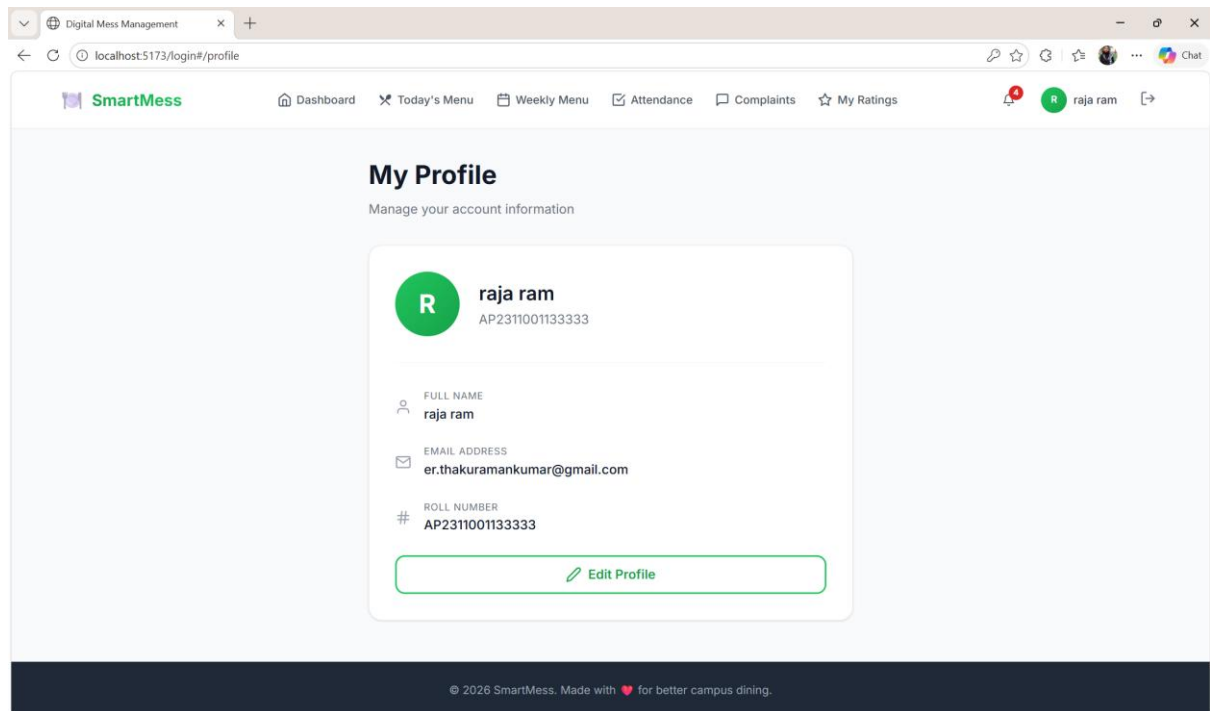
## 17. Rattings



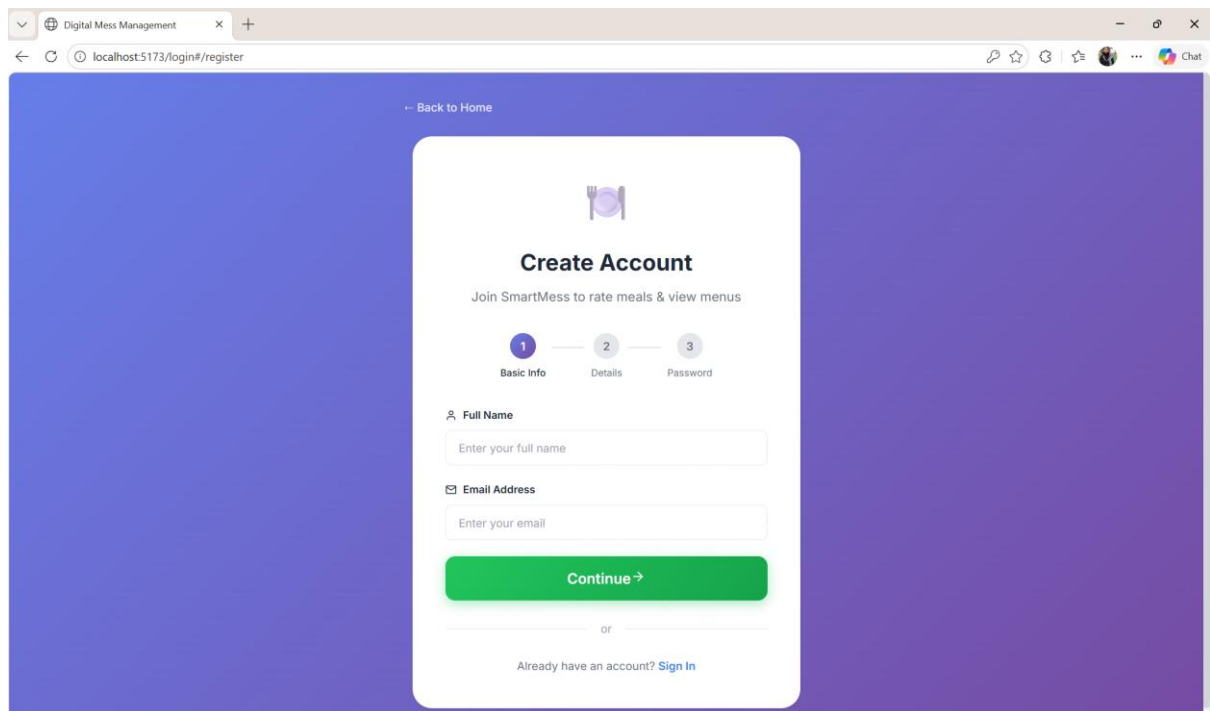
## 18. Notifications



## 19. User Profile



## 20. Create Account



## 16. FUTURE ENHANCEMENTS

Although the **Mess Management System** successfully improves the current mess operations, there are several additional features that can be implemented in the future to make the system more advanced and efficient.

One of the possible improvements is the development of a **mobile application**. A dedicated mobile app would allow users to access the system more conveniently and receive updates in real time.

Another enhancement could be the integration of a **notification system**, such as email or mobile notifications. This would help inform students about menu updates, announcements, or complaint status changes instantly.

The system can also be extended to include an **online payment feature**, allowing students to pay mess fees digitally. This would make financial transactions more convenient and reduce manual work.

An advanced **analytics module** can be added to analyze user feedback and attendance data. This would help administrators make better decisions regarding menu planning and resource management.

In addition, features like **multi-language support** can be introduced to make the system accessible to a wider range of users.

The implementation of **AI-based recommendations** could also be considered in the future. Based on previous ratings and feedback, the system can suggest menu items that are more likely to satisfy students.

Further improvements such as **QR code-based attendance**, **real-time updates**, and **improved UI design** can enhance the overall performance and user experience of the system.



## 17. CONCLUSION

The **Mess Management System** is a practical and efficient solution designed to improve the overall management of hostel mess operations. By replacing traditional manual methods with a digital platform, the system provides better organization, transparency, and ease of use for both students and administrators.

Through this project, various features such as menu management, digital attendance, feedback collection, and complaint handling have been successfully implemented. These features help in reducing manual effort, improving communication, and ensuring that all activities are managed in a structured manner.

The use of modern web technologies and a full stack development approach has enabled the system to function smoothly with proper integration between frontend, backend, and database. Security measures such as authentication and data protection further ensure that the system remains reliable and safe for users.

This project has also provided valuable learning experience in designing and developing a real-world application. It helped in understanding important concepts such as API development, database management, user authentication, and system design.

In conclusion, the Mess Management System demonstrates how technology can be effectively used to solve everyday problems. With further enhancements, it has the potential to be implemented on a larger scale and provide even more benefits in the future.